

Challenge (Habanero)

- Q1.** A park ranger recorded the number of hikers who brought water bottles and the number of hikers who did not. The two-way table is shown below. What is the relative frequency of hikers who brought water bottles and completed the trail?
- (A) 0.25
(B) 0.35
(C) 0.40
(D) 0.50
(E) 0.60
- Q2.** A car's fuel consumption is recorded in the table below. What is the probability that a car travels more than 200 miles given that it uses more than 10 gallons of fuel?
- (A) 0.30
(B) 0.40
(C) 0.50
(D) 0.60
(E) 0.70
- Q3.** The table below shows the results of a bird count. What is the association between the type of bird and its habitat?
- (A) Positive association
(B) Negative association
(C) No association
(D) Neutral association
(E) Inverse association
- Q4.** A survey of tourists is shown in the two-way table below. What percentage of tourists visited the beach and stayed in a hotel?
- (A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q5.** A two-way table is shown below. If $P(AcapB) = 0.20$, what is $P(AcupB)$?
- (A) 0.30
(B) 0.40
(C) 0.50
(D) 0.60
(E) 0.70
- Q6.** An animal shelter recorded the number of dogs and cats adopted. The two-way table is shown below. What is the relative frequency of dogs adopted by families with children?
- (A) 0.30
(B) 0.35
(C) 0.40
(D) 0.45
(E) 0.50
- Q7.** A travel company recorded the mode of transportation used by tourists. The two-way table below shows the results. What is the probability that a tourist uses a car given that they travel more than 100 miles?
- (A) 0.25
(B) 0.35
(C) 0.45
(D) 0.55
(E) 0.65
- Q8.** A park ranger recorded the number of visitors who saw a bear and the number of visitors who did not. The two-way table is shown below. What is the association between seeing a bear and the time of day?
- (A) Positive association
(B) Negative association
(C) No association
(D) Neutral association
(E) Inverse association

Open Ended Questions (FRQ)

- Q9.** A two-way table shows the results of a survey about the type of vehicle used by tourists and the type of accommodation they stayed in. The table is shown below. Calculate the relative frequency of tourists who used a car and stayed in a hotel. Show your work and explain your answer.
- Q10.** A wildlife conservation organization recorded the number of animals spotted and the type of habitat they were seen in. The two-way table is shown below. Analyze the association between the type of animal and the type of habitat. Use $x \cdot y$ to represent the product of the number of animals spotted and the number of habitats they were seen in. Calculate $x \cdot y$ and explain your answer.

Chef's Tips

- Tip 1: Ensure students understand the concept of relative frequency and its calculation.
- Tip 2: Remind students to analyze the association between variables and explain their answers.
- Tip 3: Encourage students to show their work and use proper notation when calculating $x \cdot y$.